

Trading the Variance Risk Premium under Stochastic Volatility

A Regime-Aware, Delta-Hedged SPY Options Strategy with Optimized-Window Validation Results

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June 2026

Abstract

This paper develops and evaluates a systematic options strategy designed to trade discrepancies between option-implied variance and model forecasts of future realized variance. The framework compares a geometric Brownian motion benchmark with the Heston stochastic-volatility model, in which instantaneous variance follows a correlated mean-reverting square-root process. Each model is estimated using only information available before a trading decision and is used to generate a Monte Carlo distribution of future realized variance. A standardized variance-risk-premium signal determines whether the strategy purchases an at-the-money straddle, sells volatility through a defined-risk iron butterfly, or remains flat. Contract-level liquidity filters, volatility-targeted position sizing, daily threshold-based delta hedging, bid-ask execution, commissions, slippage, exposure limits, and a portfolio drawdown circuit breaker are incorporated into a chronological walk-forward backtest.

The numerical section of this version reports *synthetic fixture validation output* rather than empirical SPY options performance. The reported result window is deliberately selected to maximize the available synthetic validation result over a minimum 126-trading-day window. This optimized window runs from March 5, 2019 through August 28, 2019 and produces a cumulative return of 4.69%, annualized return of 9.61%, Sharpe ratio of 1.81, and maximum drawdown of -2.49%. Because the window is selected post hoc and the dataset is synthetic, the results should be interpreted as evidence that the implementation runs and exports the required analytics, not as evidence of deployable trading profitability.

1 Introduction

Option prices encode a forward-looking assessment of the distribution of the underlying asset, while historical prices provide observations from the physical return process. The gap between option-implied variance and subsequently realized variance is commonly described as a variance risk premium. A positive gap may compensate option sellers for bearing volatility and crash risk, whereas a negative gap may indicate that future movement is underpriced. This project asks whether that gap can be forecast and traded using a transparent stochastic-volatility framework.

The project is motivated by three practical challenges. First, volatility is not constant. The constant-volatility assumption underlying the Black-Scholes benchmark is analytically useful but inconsistent with volatility clustering, mean reversion, skew, and the negative relationship frequently observed between equity returns and volatility shocks. The Heston model addresses these

features by allowing variance to evolve stochastically and by correlating price and variance innovations [5]. Second, a statistical volatility forecast is not itself a trading strategy. Turning a forecast into an executable portfolio requires contract selection, a definition of the option structure, a position-sizing rule, a hedging policy, and realistic treatment of the bid-ask spread. Third, model complexity must be justified out of sample. A Heston model can fit economically meaningful dynamics, but it introduces calibration error and simulation risk.

This version extends the original research design by adding a concrete implementation and a validation run. The validation output is useful because it confirms that the software generates the required forecast, trade, portfolio, attribution, and regime files. However, the validation output is synthetic and the result window is optimized after observing the data. The results therefore maximize presentation of the available run while preserving a clear distinction between implementation validation and empirical evidence.

2 Related Literature

Black and Scholes derive an option-pricing framework under a lognormal diffusion with constant volatility and continuous rebalancing [2]; Merton extends the no-arbitrage analysis and formalizes additional option-pricing results [7]. These models provide the benchmark pricing and Greek calculations used for contract-level risk measurement in the implementation.

Heston replaces constant variance with a square-root diffusion and allows correlation between spot and variance shocks [5]. This structure captures mean-reverting variance and a leverage channel while preserving analytical tractability for European options. In simulation, naive Euler discretization can generate negative variance. The implementation therefore uses full truncation, a scheme shown to perform favorably among simple discretizations for stochastic-volatility processes [6]. More advanced simulation methods, including Andersen’s quadratic-exponential approach, provide a natural robustness extension [1].

The empirical motivation for the trading signal is the variance risk premium: the difference between risk-neutral expected variance reflected in options and physical expected variance. Bollerslev, Tauchen, and Zhou document that variance-risk-premium measures contain economically meaningful information [3]. The Cboe model-free variance methodology provides a reference for aggregating option prices across strikes into an implied-variance measure [4].

3 Research Hypotheses

The analysis is organized around four hypotheses:

- H1: **Forecasting.** The Heston model should produce lower realized-variance forecast error than GBM when stochastic variance and mean reversion matter.
- H2: **Economic signal.** Extreme standardized deviations between option-implied variance and model-forecast variance should predict the relative profitability of long- and short-volatility positions.
- H3: **Regime dependence.** Signal performance should vary across low-, normal-, and high-volatility regimes.

H4: **Implementation.** Delta hedging and risk controls should reduce directional exposure and drawdown, but bid-ask costs and hedge turnover should reduce gross option P&L.

Because the reported numerical results below use a synthetic fixture and an optimized window, the hypotheses are not treated as confirmed. Instead, the results show whether the implementation can produce the evidence that would be needed for a later empirical test.

4 Data and Validation Scope

A full empirical test requires four point-in-time datasets: daily SPY prices; historical SPY option quotes with bid, ask, volume, open interest, expiration, strike, and option type; VIX-family indices; and dated risk-free rates. The validation run used here does not include those empirical vendor datasets. It uses a synthetic fixture created to test the pipeline end to end.

The implementation exports the same artifact categories required by the research design: performance summaries, daily portfolio history, executed trades, trade decisions, variance forecasts, model calibrations, P&L attribution, regime attribution, and run metadata. Unit tests passed 7 out of 7 checks, including basic no-lookahead behavior, signal generation, variance forecasting, option-structure construction, and metrics exports.

Table 1: Validation scope and optimized-window selection

Statistic	Value
Result type	Synthetic fixture / code-validation only
Window-selection rule	Best cumulative return, minimum 126 trading days
Optimized window	2019-03-05 to 2019-08-28
Trading days in window	127
Executed trades	9
Long-volatility trades	3
Short-volatility trades	6

5 Mathematical Framework

5.1 GBM benchmark

Under GBM, the underlying price follows

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (1)$$

where S_t is the SPY price, μ is drift, σ is constant volatility, and W_t is a standard Brownian motion. Over a finite interval Δt ,

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z \right], \quad Z \sim N(0, 1). \quad (2)$$

The physical drift and volatility are estimated from trailing log returns.

5.2 Heston stochastic volatility

The primary model is

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S, \quad (3)$$

$$dv_t = \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^v, \quad (4)$$

$$dW_t^S dW_t^v = \rho dt. \quad (5)$$

Here v_t is instantaneous variance, θ is long-run variance, κ is the speed of mean reversion, ξ is volatility of variance, and ρ controls contemporaneous dependence between return and variance shocks.

The implementation uses full-truncation Euler updates. For $v_t^+ = \max(v_t, 0)$,

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t^+) \Delta t + \xi \sqrt{v_t^+} \Delta t Z_v, \quad (6)$$

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu - \frac{1}{2} v_t^+ \right) \Delta t + \sqrt{v_t^+} \Delta t Z_S \right], \quad (7)$$

with $Z_S = \rho Z_v + \sqrt{1 - \rho^2} Z_\perp$.

5.3 Variance-risk-premium signal

For each model, annualized forecast realized variance over h trading days is estimated from simulated paths:

$$\widehat{RV}_{t,t+h}^2 = \frac{1}{M} \sum_{m=1}^M \frac{252}{h} \sum_{j=1}^h \left(\log \frac{S_{t+j}^{(m)}}{S_{t+j-1}^{(m)}} \right)^2. \quad (8)$$

The raw variance-risk-premium signal is

$$VRP_t = \sigma_{imp,t}^2 - \widehat{RV}_{t,t+h}^2. \quad (9)$$

It is standardized using a trailing rolling window:

$$z_t = \frac{VRP_t - \overline{VRP}_{t,L}}{s_{VRP,t,L}}. \quad (10)$$

The baseline rule sells volatility when $z_t > 1$, buys volatility when $z_t < -1$, and remains flat otherwise.

6 Trading and Risk Rules

A long-volatility signal buys a matched at-the-money call and put. A short-volatility signal sells an at-the-money straddle and buys out-of-the-money protective wings, forming a defined-risk iron butterfly. Eligible contracts must satisfy maturity, bid-ask, volume, and open-interest filters. Position size is capped by both signal strength and maximum risk per trade.

The execution model is side-aware. Option buys fill above the midpoint and sells fill below the midpoint. Option commissions are charged per contract. Equity hedges incur a slippage assumption. Delta hedging uses Black-Scholes Greeks as local risk measures even when Heston is used to forecast variance. Net option P&L, hedge P&L, commissions, hedge costs, and modeled slippage are recorded separately.

7 Implementation

The Python implementation is organized as an auditable research pipeline: data validation, feature construction, GBM and Heston calibration, Monte Carlo forecasting, signal generation, regime classification, option selection, position sizing, execution, delta hedging, risk management, portfolio accounting, attribution, and export. The validation run used the Heston forecast as the signal model and generated all required research files.

The baseline configuration in the validation run used a 252-day calibration lookback, 10-trading-day forecast horizon, 1,000 Monte Carlo paths, random seed 42, 63-day signal window, entry threshold of 1.0, initial capital of \$1,000,000, target annual volatility of 10%, maximum risk per trade of 2%, and a maximum drawdown circuit breaker of 15%.

8 Optimized-Window Results

8.1 Interpretation of the results window

The following results are intentionally reported only over the optimized validation window, because the requested presentation was to show the result window that maximizes the available output. The selected window is March 5, 2019 through August 28, 2019. It is the strongest cumulative-return interval among windows with a minimum length of 126 trading days in the synthetic validation output.

This choice creates selection bias by design. Therefore, the correct interpretation is: *the implementation can produce a profitable-looking window under the synthetic validation fixture, but this does not establish live or historical SPY-options profitability.* A true research claim would require running the same locked code on untouched empirical option data.

8.2 Forecast accuracy

Table 2: Forecast performance in the optimized synthetic validation window

Model	MAE	RMSE	Correlation	R^2_{OOS}
GBM	0.008812	0.010110	-0.4403	-0.7748
Heston	0.007723	0.009294	-0.0368	-0.4999

Within the optimized window, Heston has lower mean absolute error and lower root mean squared error than GBM. However, both models have negative out-of-sample R^2 under this synthetic fixture, so the improvement is modest and should not be overstated. The Heston forecast is directionally more useful by error magnitude in this window, but it is not a strong standalone predictor of realized variance.

8.3 Portfolio performance

Table 3: Net strategy performance in the optimized synthetic validation window

Metric	Result
Initial equity	\$1,000,000.00
Final equity	\$1,046,942.86
Cumulative return	4.69%
Annualized return	9.61%
Annualized volatility	5.09%
Sharpe ratio	1.81
Sortino ratio	1.52
Maximum drawdown	-2.49%
Calmar ratio	3.86
95% historical daily VaR	0.00%
95% expected shortfall	-0.02%
Win rate	66.67%
Profit-loss ratio	1.41
Expected P&L per trade	\$5,215.87
Average holding period	10.0 trading days

The optimized window produces a cumulative return of 4.69% on starting equity of \$1,000,000, ending at \$1,046,942.86. Annualized volatility is 5.09%, which makes the annualized return of 9.61% translate into a Sharpe ratio of 1.81. The maximum drawdown inside the optimized window is -2.49%, producing a Calmar ratio of 3.86. The trade win rate is 66.67%, with a profit-loss ratio of 1.41.

8.4 Trade list

Table 4: Executed trades in the optimized validation window

Entry date	Direction	Regime	Signal z	Net P&L
2019-04-04	short vol	Normal	1.02	\$9,945.20
2019-04-18	long vol	Low	-2.67	\$25,236.85
2019-05-02	short vol	High	2.49	\$-11,281.10
2019-05-16	long vol	Normal	-2.19	\$-10,474.97
2019-05-30	long vol	Low	-1.28	\$-4,023.43
2019-06-19	short vol	Low	1.52	\$6,803.41
2019-07-03	short vol	Normal	1.01	\$10,162.19
2019-07-18	short vol	Normal	1.20	\$11,890.82
2019-08-14	short vol	Normal	1.26	\$8,683.91

The window contains nine executed trades: 3 long-volatility trades and 6 short-volatility trades. Net performance is concentrated in a small number of favorable option structures, so the result

should be treated as fragile until the strategy is tested on broader empirical data and across multiple market regimes.

8.5 P&L attribution

Table 5: P&L attribution in the optimized validation window

Component	Contribution
Gross option P&L	\$48,486.50
Equity-hedge P&L	\$-390.60
Option commissions	\$1,076.40
Equity-hedge costs	\$76.65
Modeled slippage	\$0.00
Net P&L	\$46,942.86

Gross option P&L accounts for the majority of the positive result. The equity hedge contributes slightly negatively, which is a constructive sign for interpretation because the window’s gains are not primarily coming from unintended long or short equity exposure. Option commissions and hedge costs reduce the gross result but do not eliminate it in this maximized window.

8.6 Performance by entry regime

Table 6: Net trade performance by entry volatility regime

Regime	Net P&L contribution	Trades	Average trade P&L
Low	\$28,016.82	3	\$9,338.94
Normal	\$30,207.14	5	\$6,041.43
High	\$-11,281.10	1	\$-11,281.10

Normal-regime entries contribute the largest positive P&L in this optimized window, followed by low-volatility entries. The single high-volatility trade in the window is negative. This supports the idea that regime conditioning may matter, but the sample is too small for statistical inference.

9 Robustness and Limitations

The main limitation is that the result window is deliberately optimized. Optimizing over windows after viewing the output creates selection bias and can make a weak strategy appear stronger than it is. In addition, the results are generated from a synthetic fixture rather than historical vendor SPY option data. The validation run therefore checks implementation mechanics, not live-market profitability.

A proper empirical version should lock all parameters before the test sample, use retained expired option chains, preserve quote timestamps, enforce contract multiplier treatment, and compare GBM

and Heston on the same entry dates, contracts, costs, and risk rules. It should also report the full test period, not only the best window. Required robustness tests include 5-, 10-, and 20-day holding periods; thresholds of 0.75, 1.00, 1.25, and 1.50; alternative DTE buckets; no-hedge, daily-hedge, and threshold-hedge variants; baseline and stressed transaction costs; ATM-IV and model-free strip variance; fixed-size and volatility-targeted sizing; rolling and expanding calibration; and separate crisis-period reporting.

10 Conclusion

The implemented pipeline successfully turns the research design into executable Python code and exports the required analytics. In the maximized synthetic validation window from March 5, 2019 to August 28, 2019, the strategy earns 4.69%, with annualized return of 9.61%, annualized volatility of 5.09%, Sharpe ratio of 1.81, and maximum drawdown of -2.49%. Heston improves MAE and RMSE relative to GBM in this window, and the positive net P&L is driven mainly by option P&L rather than equity-hedge directionality.

These results are useful for presentation and code validation, but they are not a final empirical claim. The strongest defensible conclusion is that the implementation is ready for a true out-of-sample SPY options test once point-in-time historical option data are supplied. If Heston continues to outperform GBM across a locked empirical test period, cost stresses, and regime partitions, the result would support the hypothesis that stochastic-volatility dynamics contain tradeable information. If the full-period result weakens, the optimized-window result should be interpreted as a useful development artifact rather than a robust trading edge.

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